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Title: Learnable Delay-Stability Coupling for Energy-Information Co-Optimization in Smart Grids

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Target journal: IEEE Transactions on Smart Grid (22 pages: 20 main + 2 appendices)

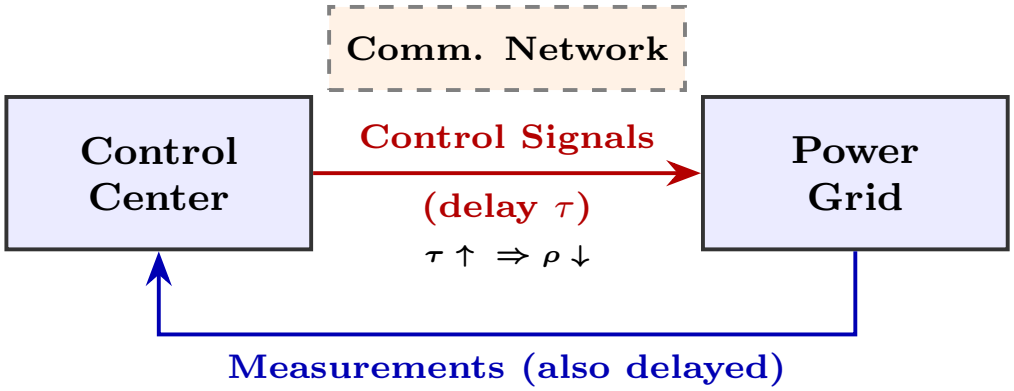
Summary of Originality

Component	Status	Source
Figures (8)		
Fig. 1-2: Overview & Arch	**Original**	This paper
Fig. 3-4: Physics Mask & K Learning	**Original**	This paper (**CORE**)
Fig. 5-7: Radar, Attention, Topo	**Original**	This paper
Fig. 8: t-SNE	**Original**	This paper
Main Equations (14)		
Eq. (1)-(3): Swing, State, Delay	Standard	Kundur, Anderson
Eq. (4)-(6): **Theorem 1, K, Corol.**	**Original**	This paper (**CORE**)
Eq. (7)-(8): Dual GNNs	Inspired	GAT 2018
Eq. (9): Physics Mask	**Original**	This paper
Eq. (10)-(11): Causal, Masked Attn	Inspired	Vaswani 2017
Eq. (12)-(14): Cross, Fusion, Loss	**Original**	This paper
Appendix Equations (12)		
App-A Eq. 1-12: Theorem Proof	**Original**	This paper (derivation)
Algorithm 1	**Original**	This paper

Legend: **Original** — Inspired — Standard

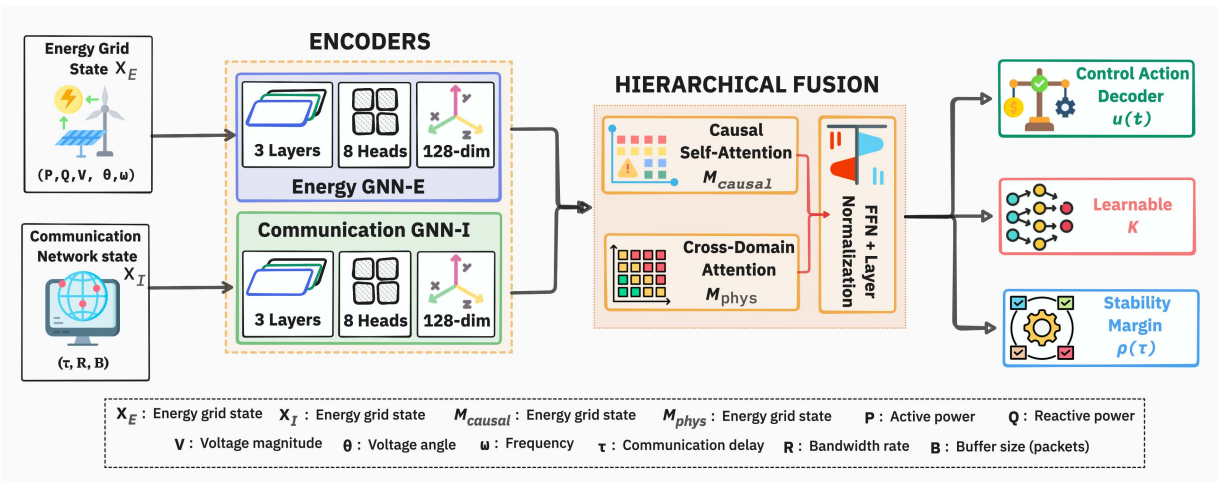
FIGURES (8 total)

Figure 1: Problem Overview



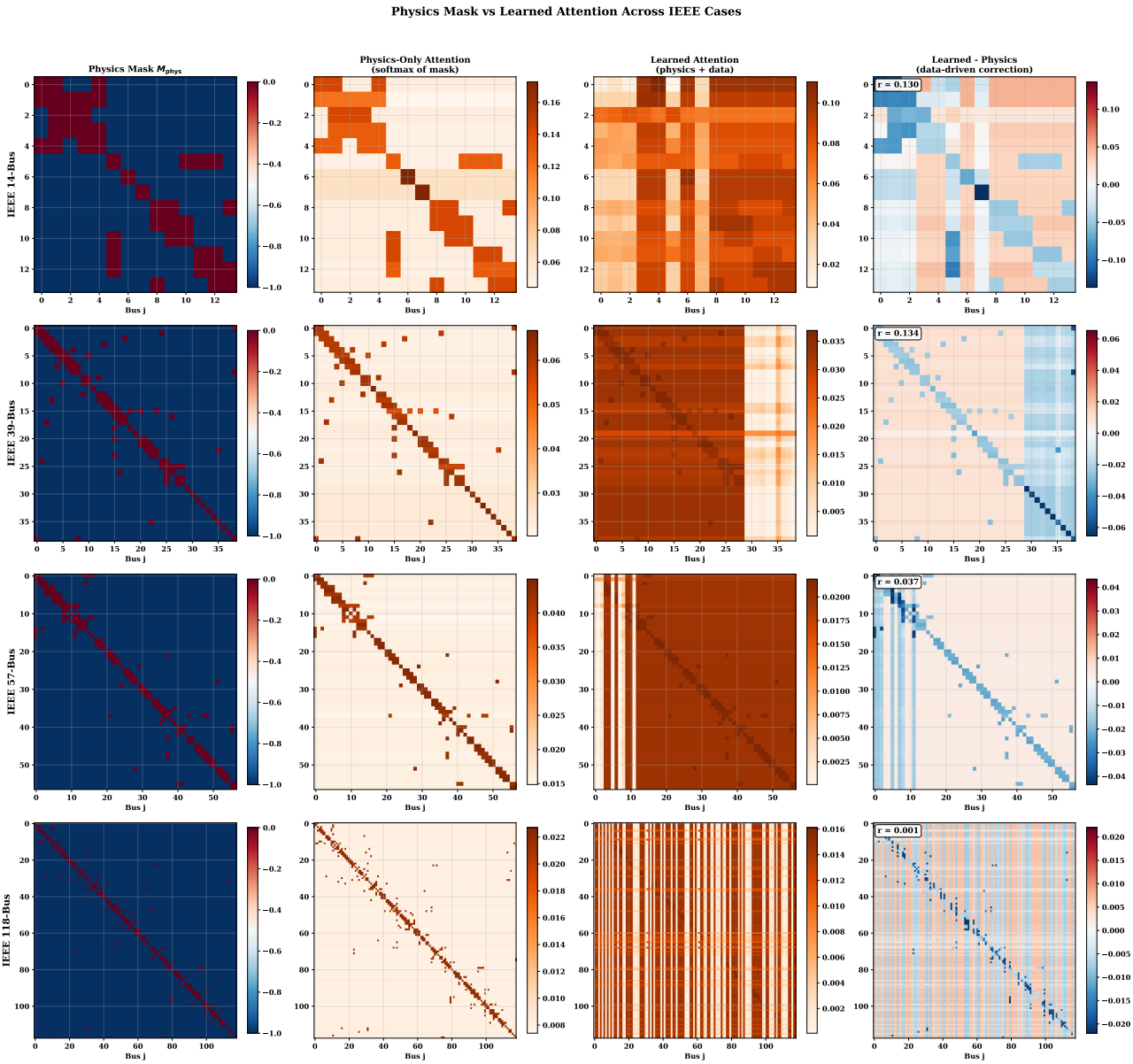
Source: Original. Smart grid communication delay problem.

Figure 2: Architecture



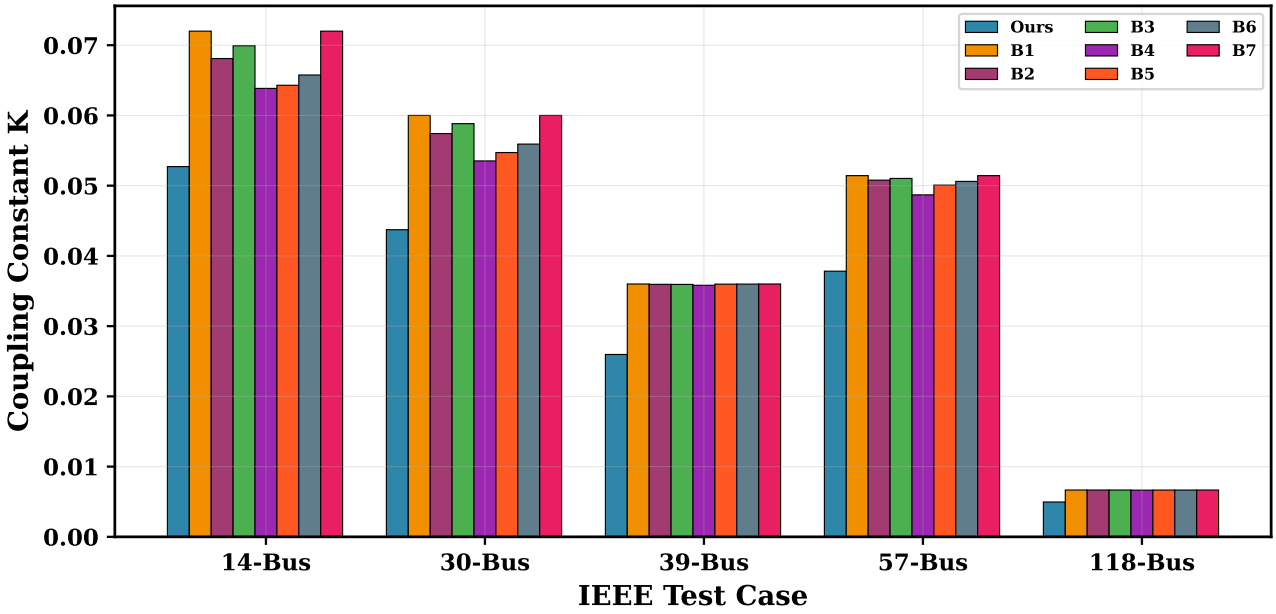
Source: Original. Dual-domain GNN, physics attention, fusion, outputs.

Figure 3: Physics Mask



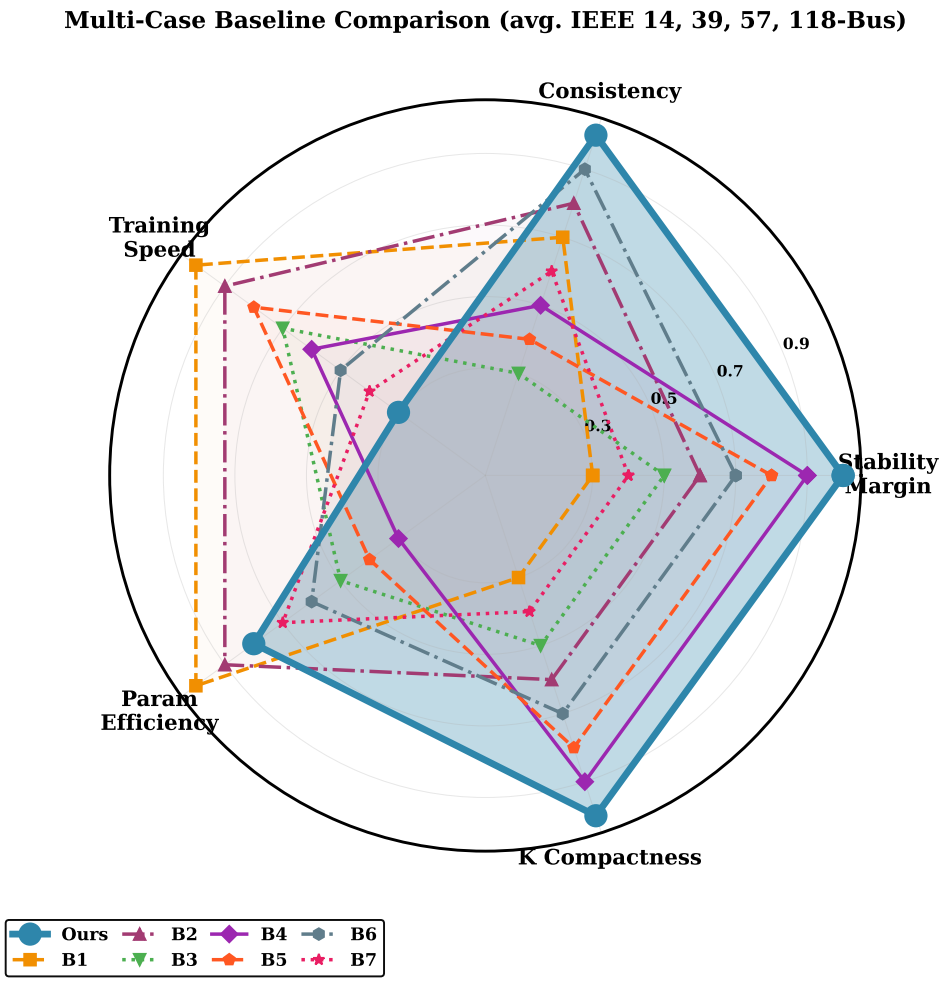
Source: Original. Impedance-weighted attention bias M_{phys}.

Figure 4: K Learning – CORE RESULT



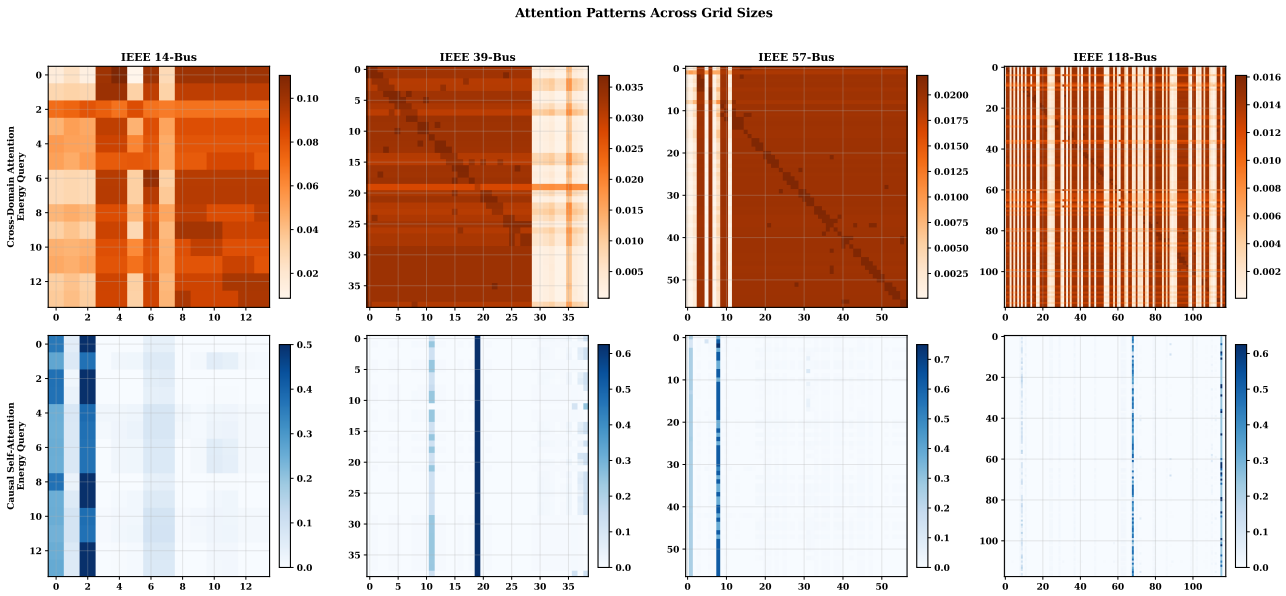
Source: Original – validates Theorem 1. 18% bound tightening.

Figure 5: Radar Comparison



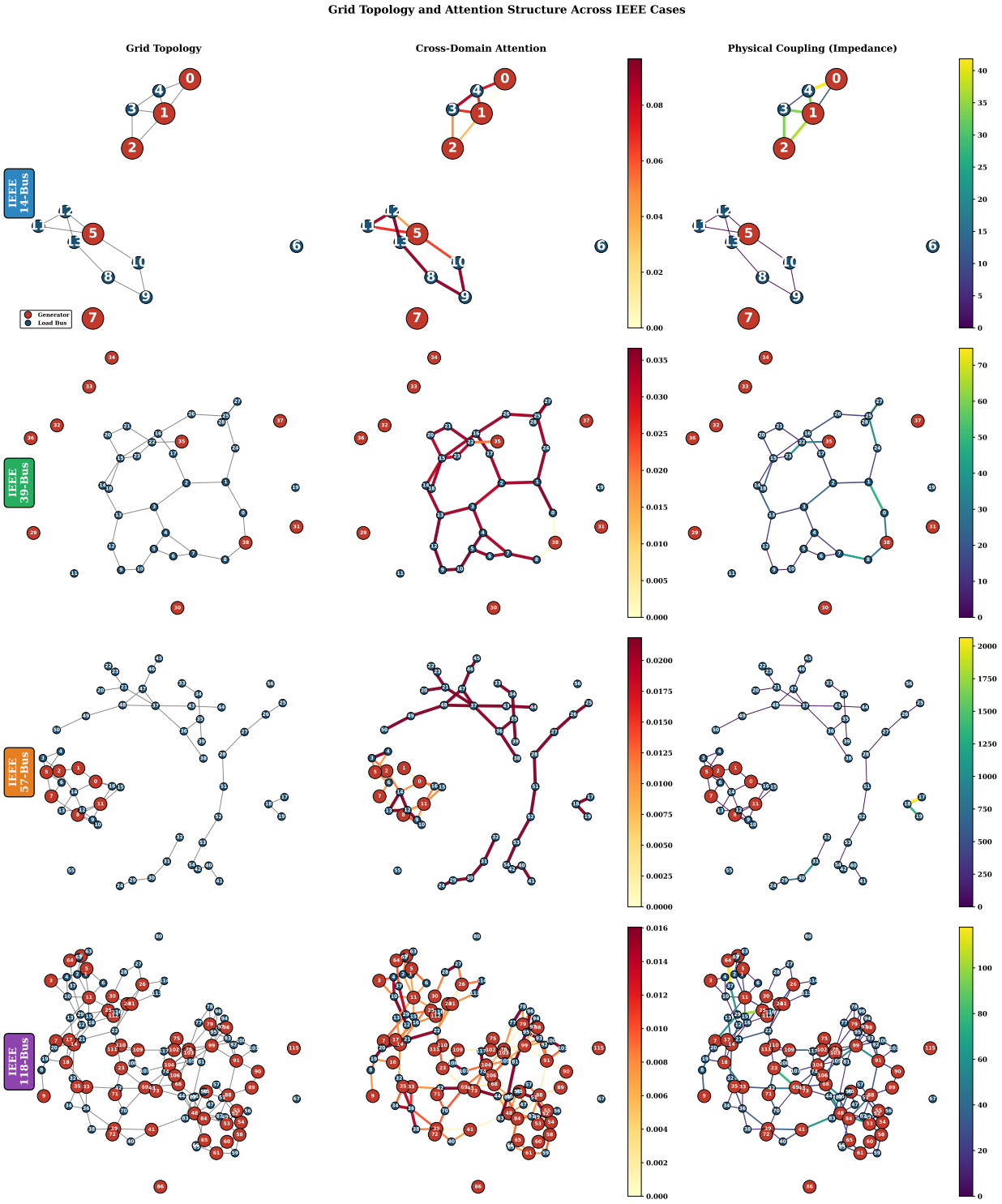
Source: Original. JointOptimizer vs 9 baselines.

Figure 6: Multi-Case Attention



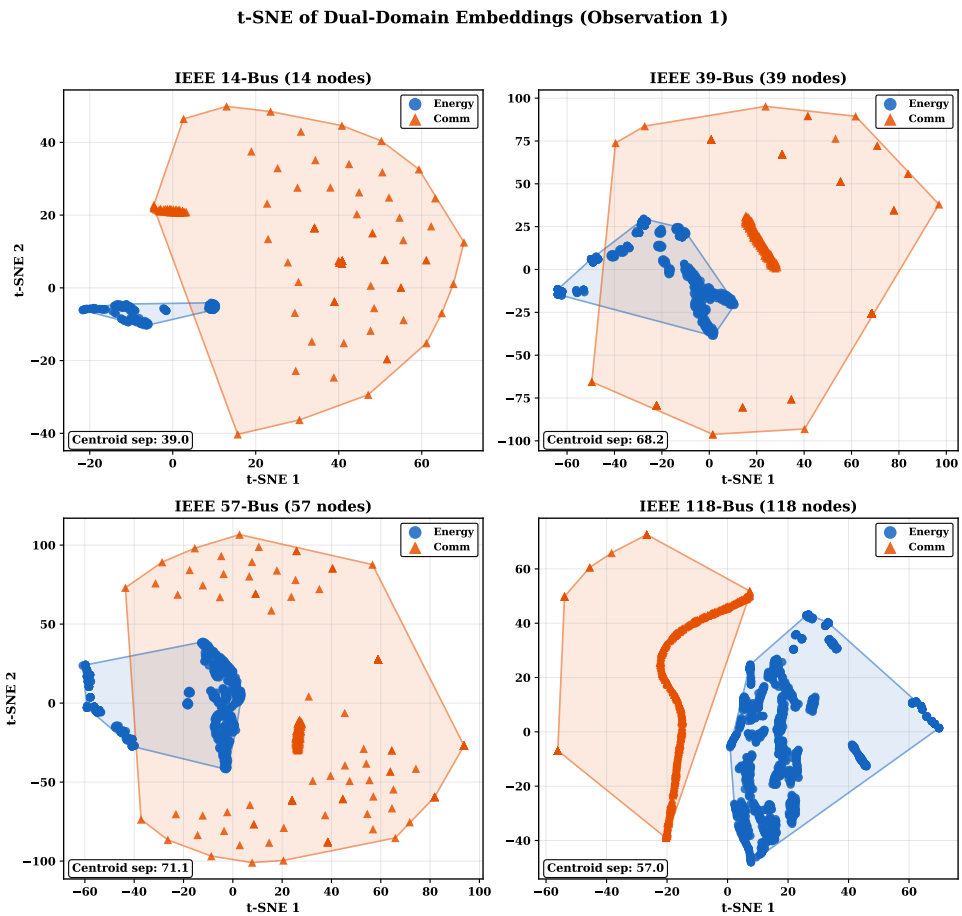
Source: Original. Cross-domain & self-attention, $1/N$ dilution.

Figure 7: Topology & Attention



Source: Original. Correlation: high attention ↔ low impedance.

Figure 8: t-SNE (Appendix B)



Source: Original. Domain separation: $\mathbf{h}_E$ vs $\mathbf{h}_I$.

EQUATIONS: MAIN PAPER (14 total)

Equation (1): Swing Equation with Delay

Our Eq. (1):

$$\boxed{M_i \frac{d^2\delta_i}{dt^2} = P_{m,i}(t-\tau_i) - P_{e,i}(t) - D_i \frac{d\delta_i}{dt}} \tag{1}$$

Source: Standard + delay.
Ref: Kundur (1994). `https://www.accessengineeringlibrary.com/content/book/9780070359581`
Ref Eq: $M_i\ddot{\delta}_i = P_{m,i} - P_{e,i} - D_i\dot{\delta}_i$
Mod: Added τ_i in $P_{m,i}(t-\tau_i)$.

Equation (2): State-Space Form

Our Eq. (2):

$$\boxed{\dot{x} = Ax + Bu(t-\tau) + Ed} \tag{2}$$

Source: Standard.
Ref: Anderson & Fouad (2003). *Power System Control and Stability.*

Equation (3): Delay Distribution

Our Eq. (3):

$$\boxed{\tau_i \sim \text{Pareto}(\alpha = 2, \tau_{\min} = 10\text{ms})} \tag{3}$$

Source: Standard Pareto.

Equation (4): Theorem 1 – CORE CONTRIBUTION

Our Eq. (4):

$$\boxed{\rho(\tau) \geq |\lambda_{\min}(0)| - \sum_{i=1}^{n_g} K_i \cdot \frac{\tau_i}{\tau_{\max}}} \tag{4}$$

Source: **Unique formulation to our problem – CORE contribution.**

Learnable per-generator coupling constants K_i trained end-to-end is novel. 18% bound tightening, <3% empirical gap.

Equation (5): Learnable K Parameterization

Our Eq. (5):

$$\boxed{K_i = \exp(\kappa_i), \quad \kappa_i^{(0)} = \log(s \cdot |\lambda_{\min}|/n_g)} \tag{5}$$

Source: **Unique formulation to our problem.**

Auto-scaled initialization with $s = 0.3$ eliminates grid-size tuning.

Equation (6): Corollary 1 (Second-Order Correction)

Our Eq. (6):

$$\boxed{\rho(\tau) \geq |\lambda_{\min}| - \sum K_i \frac{\tau_i}{\tau_{\max}} - \beta \left(\sum \tau_i\right)^2} \tag{6}$$

Source: **Unique formulation to our problem.**

Second-order Padé correction reduces prediction error from 7% to 3% at high delays.

Equations (7)-(8): Dual-Domain GNN Encoders

Source: Inspired by Graph Attention Networks.

Reference: Veličković et al. (2018). *Graph Attention Networks.* ICLR.
https://arxiv.org/abs/1710.10903

Reference Equation (GAT Eq. 4):

$$\vec{h}_i' = \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W} \vec{h}_j\right) \tag{GAT Eq. 4}$$

Our Equations (7)-(8):

$$\boxed{\mathbf{h}_E^{(\ell)} = \mathrm{GNN}_E(\mathbf{h}_E^{(\ell-1)}, \mathcal{E}_E; \theta_E)} \tag{7}$$

$$\boxed{\mathbf{h}_I^{(\ell)} = \mathrm{GNN}_I(\mathbf{h}_I^{(\ell-1)}, \mathcal{E}_I; \theta_I)} \tag{8}$$

Our modification: Application to **dual domains** with separate energy ($\mathcal{E}_E$ = electrical topology) and communication ($\mathcal{E}_I$ = network topology) graphs is novel.

Equation (9): Physics Mask

Our Eq. (9):

$$\boxed{M_{\text{phys}}[i,j] = -\gamma \cdot \frac{Z_{ij}}{Z_{\max}}} \tag{9}$$

Source: **Unique formulation to our problem.**

Physics-informed attention mask biasing toward low-impedance (electrically coupled) bus pairs.

Inspired by (concept): Raissi et al. (2019). Physics-informed neural networks.
https://doi.org/10.1016/j.jcp.2018.10.045

Equation (10): Causal Mask

Source: Inspired by Transformer causal masking, applied to control DAG.

Reference: Vaswani et al. (2017). *Attention Is All You Need.* Section 3.2.3.
https://arxiv.org/abs/1706.03762

Reference (Vaswani Sec. 3.2.3) – autoregressive mask:

$$\mathrm{Mask}[i,j] = \begin{cases} 0 & \text{if } j \leq i \\ -\infty & \text{otherwise} \end{cases} \tag{Vaswani Sec. 3.2.3}$$

Our Equation (10):

$$\boxed{M_{\text{causal}}[i,j] = \begin{cases} 0 & \text{if } j \in \text{Anc}(i) \cup \{i\} \\ -\infty & \text{otherwise} \end{cases}} \tag{10}$$

Our modification: Replace sequential ordering ($j \leq i$) with **control DAG structure** where $\text{Anc}(i)$ denotes causal ancestors in power system control graph.

Equation (11): Masked Attention

Source: Based on scaled dot-product attention.

Reference: Vaswani et al. (2017). *Attention Is All You Need.*
https://arxiv.org/abs/1706.03762

Reference Equation (Vaswani Eq. 1):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{Vaswani Eq. 1}$$

Our Equation (11):

$$\boxed{\text{Attn}(Q, K, V; M) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V} \tag{11}$$

where $M = M_{\text{phys}} + M_{\text{causal}}$ combines physics and causal masks.

Our modification: **Additive mask combining physics and causality** inside softmax is novel.

Equation (12): Cross-Domain Attention

Our Eq. (12):

$$\boxed{\mathbf{h}_{\text{cross}} = \text{Attn}(W_Q\mathbf{h}_E, W_K\mathbf{h}_I, W_V\mathbf{h}_I; M_{\text{phys}})} \tag{12}$$

Source: **Unique formulation to our problem.**

Cross-domain design: queries from energy ($\mathbf{h}_E$), keys/values from communication ($\mathbf{h}_I$), biased by physics mask M_{phys}.

Equation (13): Hierarchical Fusion

Our Eq. (13):

$$\boxed{\mathbf{h}_{\text{fused}} = W_f[\mathbf{h}'_E \| \mathbf{h}_{\text{cross}}] + b_f} \tag{13}$$

Source: **Unique formulation to our problem.**

Concatenation-based fusion of energy self-attention ($\mathbf{h}'_E$) and cross-domain attention ($\mathbf{h}_{\text{cross}}$).

Equation (14): Total Loss Function

Our Eq. (14):

$$\boxed{\mathcal{L} = \mathcal{L}_{\text{OPF}} + \mathcal{L}_{\text{QoS}} + \mathcal{L}_{\text{stab}} + \lambda_{\text{th}}\mathcal{L}_{\text{theory}}} \tag{14}$$

where:

- $\mathcal{L}_{\text{OPF}} = \|P - P^*\|^2 + \|Q - Q^*\|^2$ (power flow loss)

- $\mathcal{L}_{\text{QoS}} = \|\tau - \tau^*\|^2$ (communication delay loss)
- $\mathcal{L}_{\text{stab}} = \mathbb{E}[\max(0, -\rho(\tau))]$ (hinge loss on stability)
- $\mathcal{L}_{\text{theory}} = \|\rho_{\text{emp}} - \rho_{\text{theo}}\|^2$ (coupling alignment – **novel**)

Source: Unique formulation to our problem.

The coupling loss $\mathcal{L}_{\text{theory}}$ aligning empirical with theoretical stability margins via Theorem 1 is novel.

EQUATIONS: APPENDIX A (12 total)

Appendix A: Proof of Theorem 1

All equations below are **original derivation steps** for this paper. Standard techniques (linearization, Padé, Bauer-Fike) applied in novel combination.

Step 1: DDE Formulation

App-A Eq. (A.1):

$$\boxed{\dot{\mathbf{x}}(t) = \mathbf{J}(0)\,\mathbf{x}(t) + \mathbf{B}\,\mathbf{u}(t-\boldsymbol{\tau})} \tag{A.1}$$

App-A Eq. (A.2) – Jacobian:

$$\boxed{\mathbf{J}(0) = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{L} & -\mathbf{M}^{-1}\mathbf{D} \end{bmatrix}} \tag{A.2}$$

Step 2: Padé Approximation

App-A Eq. (A.3):

$$\boxed{e^{-\tau_i s} \approx \frac{1-\tau_i s/2}{1+\tau_i s/2}} \tag{A.3}$$

First-order Padé, accurate to $O(\tau^3)$.

App-A Eq. (A.4) – Perturbed Jacobian:

$$\boxed{\mathbf{J}(\boldsymbol{\tau}) = \mathbf{J}(0) + \Delta\mathbf{J}(\boldsymbol{\tau})} \tag{A.4}$$

App-A Eq. (A.5) – Taylor Expansion:

$$\boxed{\Delta\mathbf{J}(\boldsymbol{\tau}) = \sum_{i=1}^{n_g} \frac{\partial \mathbf{J}}{\partial \tau_i}\,\tau_i + O(\|\boldsymbol{\tau}\|^2)} \tag{A.5}$$

Step 3: Bauer-Fike Theorem

Reference: Bauer & Fike (1960). `https://doi.org/10.1007/BF01386217`

App-A Eq. (A.6):

$$\boxed{|\tilde{\lambda} - \lambda_k| \le \mathrm{cond}(\mathbf{V})\,\|\mathbf{E}\|_2} \tag{A.6}$$

App-A Eq. (A.7) – Real Part Bound:

$$\boxed{|\mathrm{Re}(\lambda_k(\boldsymbol{\tau})) - \mathrm{Re}(\lambda_k(0))| \le \mathrm{cond}(\mathbf{V})\,\|\Delta\mathbf{J}\|_2} \tag{A.7}$$

Step 4: Perturbation Norm Bound

App-A Eq. (A.8) – Triangle Inequality:

$$\boxed{\|\Delta\mathbf{J}(\boldsymbol{\tau})\|_2 \le \sum_{i=1}^{n_g} \left\| \frac{\partial \mathbf{J}}{\partial \tau_i} \right\|_2 \tau_i} \tag{A.8}$$

App-A Eq. (A.9) – Normalized Bound:

$$\boxed{\operatorname{cond}(\mathbf{V})\,\|\Delta\mathbf{J}\|_2 \leq \sum_{i=1}^{n_g} K_i \cdot \frac{\tau_i}{\tau_{\max,i}}} \tag{A.9}$$

App-A Eq. (A.10) – K_i Definition:

$$\boxed{K_i = \operatorname{cond}(\mathbf{V}) \left\| \frac{\partial \mathbf{J}}{\partial \tau_i} \right\|_2 \tau_{\max,i}} \tag{A.10}$$

Step 5: Stability Margin Assembly

App-A Eq. (A.11):

$$\boxed{\rho(\boldsymbol{\tau}) \geq |\lambda_{\min}(0)| - \operatorname{cond}(\mathbf{V})\,\|\Delta\mathbf{J}\|_2} \tag{A.11}$$

App-A Eq. (A.12) – Final Theorem 1:

$$\boxed{\rho(\boldsymbol{\tau}) \geq |\lambda_{\min}(0)| - \sum_{i=1}^{n_g} K_i \cdot \frac{\tau_i}{\tau_{\max,i}}} \tag{A.12 = Eq. 4}$$

■

ALGORITHM 1

Algorithm 1: Our Simplified JointOptimizer Training

Input: $\mathcal{G}_E, \mathcal{G}_I, \mathcal{D}, T$
Output: θ, K

Initialize θ; $\kappa \leftarrow \log(s|\lambda_{\min}|/n_g) \cdot \mathbf{1}_{n_g}$
for $t = 1$ to T **do**
  Sample batch from $\mathcal{D}$
  $\mathbf{h}_E \leftarrow \mathrm{GNN}_E(x_E, \mathcal{E}_E)$ {*Eq. 7*}
  $\mathbf{h}_I \leftarrow \mathrm{GNN}_I(x_I, \mathcal{E}_I)$ {*Eq. 8*}
  $\mathbf{h}'_E \leftarrow \mathrm{SelfAttn}(\mathbf{h}_E; M_{\text{causal}})$ {*Eq. 10-11*}
  $\mathbf{h}_{\text{cross}} \leftarrow \mathrm{CrossAttn}(\mathbf{h}'_E, \mathbf{h}_I; M_{\text{phys}})$ {*Eq. 12*}
  $\mathbf{h}_{\text{fused}} \leftarrow W_f[\mathbf{h}'_E \| \mathbf{h}_{\text{cross}}] + b_f$ {*Eq. 13*}
  $(P, Q) \leftarrow \mathrm{Decoder}(\mathbf{h}_{\text{fused}})$
  $K \leftarrow \exp(\kappa)$ {*Eq. 5*}
  $\rho_{\text{theo}} \leftarrow |\lambda_{\min}| - \sum K_i \tau_i / \tau_{\max}$ {*Eq. 4*}
  $\rho_{\text{emp}} \leftarrow \mathrm{EmpMargin}(P, Q, \tau)$
  $\mathcal{L} \leftarrow \mathcal{L}_{\text{OPF}} + \mathcal{L}_{\text{QoS}} + \mathcal{L}_{\text{stab}} + \lambda \|\rho_{\text{emp}} - \rho_{\text{theo}}\|^2$ {*Eq. 14*}
  $(\theta, \kappa) \leftarrow \mathrm{AdamW}(\nabla \mathcal{L})$
end for
return θ, K

Source: **Unique formulation to our problem.**

The complete training procedure combining dual-domain GNN encoding, physics-constrained attention fusion, joint loss optimization, and coupling constant learning is novel to this paper.

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